JEE–MAIN EXAMINATION – JANUARY 2026

(HELD ON WEDNESDAY 21st JANUARY 2026) **TIME : 3:00 PM TO 6 :00 PM**

CHEMISTRY

TEST PAPER WITH ANSWER

SECTION-A

51. Consider the following spectral lines for atomic hydrogen:

A. First line of Paschen series

B. Second line of Balmer series

C. Third line of Paschen series

D. Fourth line of Bracket sereis

The correct arrangement of the above lines in ascending order of energy is:

(1) D < C < A < B

(2) A < B < C < D

(3) C < D < B < A

(4) D < A < C < B

52. Match **List-I** with **List-II**.

	List-I Pair of Compounds		**List-II Type of Isomers**
A.	2-Methylpropene and but-1-ene	I.	Stereoisomers
B.	Cis-but-2-ene and trans-but-2-ene	II.	Position isomers
C.	2-Butanol and diethyl ether	III.	Chain isomers
D.	But-1-ene and but-2-ene	IV.	Functional group isomers

Choose the **correct** answer from the options given below:

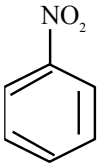
(1) A-III, B-I, C-IV, D-II

(2) A-III, B-I, C-II, D-IV

(3) A-I, B-IV, C-III, D-II

(4) A-II, B-I, C-IV, D-III

53.



(1) $Br_2/FeBr_3/\Delta$
(2) $Sn/HCl/\Delta$
(3) pH neutralisation
(4) Br_2/H_2O
(5) $NaNO_2/HBr$, 0–5°C
(6) CuBr/NaBr

→ Major Product (P)

Consider the above sequence of reactions. The number of bromine atom(s) in the final product (P) will be:

(1) 1 (2) 6 (3) 5 (4) 3

54. Aqueous HCl reacts with $MnO_2(s)$ to form $MnCl_2(aq)$, $Cl_2(g)$ and $H_2O(l)$. What is the weight (in g) of Cl_2 liberated when 8.7 g of $MnO_2(s)$ is reacted with excess aqueous HCl solution ? (Given Molar mass in g mol^{-1} Mn = 55, Cl = 35.5, O = 16, H = 1)

(1) 7.1 (2) 71 (3) 21.3 (4) 14.2

55. By usual analysis, 1.00g of compound (X) gave 1.79g of magnesium pyrophosphate. The percentage of phosphorus in compound (X) is: (nearest integer)

(Given, molar mass in g mol^{-1}: O = 16, Mg = 24, P = 31)

(1) 50 (2) 30 (3) 20 (4) 40

56. Consider the following data:

$\Delta_f H^{\ominus}$(methane, g) = –X kJ mol^{-1}

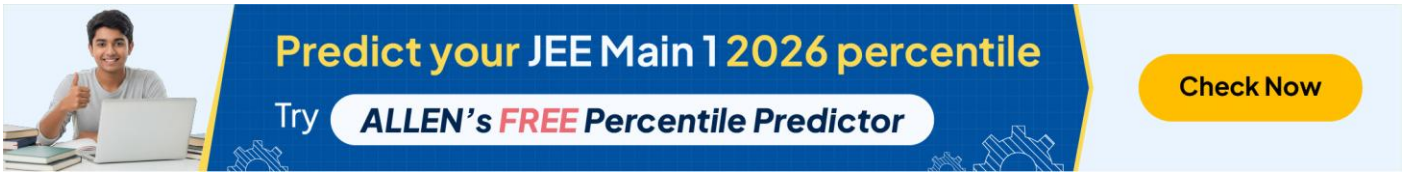
Enthalpy of sublimation of graphite = Y kJ mol^{-1}

Dissociation enthalpy of H_2 = Z kJ mol^{-1}

The bond enthalpy of C – H bond is given by:

(1) $\frac{X+Y+2Z}{4}$ (2) $\frac{X+Y+4Z}{2}$

(3) X + Y + Z (4) $\frac{-X+Y+Z}{4}$



57. Match **List-I** with **List-II**.

	List-I Reagents		List-II Reaction Name (Involving aldehydes)
A.	H_2, Pd-$BaSO_4$	I.	Etard Reaction
B.	$SnCl_2$, HCl	II.	Rosenmund Reduction
C.	CrO_2Cl_2, CS_2	III.	Gatterman–Koch Reaction
D.	CO,HCl, Anhyd. $AlCl_3$	IV.	Stephen Reaction

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I

(2) A-IV, B-III, C-I, D-II

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-IV, C-I, D-III

58. Decomposition of A is a first order reaction at T(K) and is given by A(g) → B(g) + C(g). In a closed 1 L vessel, 1 bar A(g) is allowed to decompose at T(K). After 100 minutes, the total pressure was 1.5 bar. What is the rate constant (in min^{-1}) of the reaction ? (log 2 = 0.3)

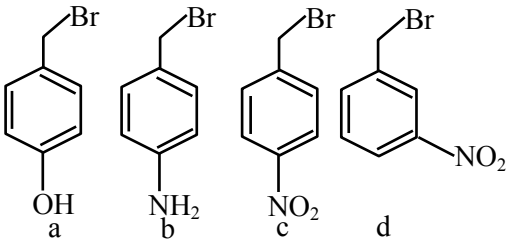
(1) 6.9×10^{-1}

(2) 6.9×10^{-3}

(3) 6.9×10^{-2}

(4) 6.9×10^{-4}

59. The correct order of reactivity of the following benzyl halides towards reaction with KCN is:



(1) a > b > c > d

(2) b > a > d > c

(3) b > a > c > d

(4) a > b > d > d

60. Given below are two statements:

Statement-I: The correct order in terms of atomic/ionic radii is Al > Mg > Mg^{2+} > Al^{3+}.

Statement-II: The correct order in terms of the magnitude of electron gain enthalpy is Cl > Br > S > O.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is false but **Statement II** is true

(3) **Statement I** is true but **Statement II** is false

(4) Both **Statement I** and **Statement II** are true

61. The **correct** statements are:

A. Activation energy for enzyme catalysed hydrolysis of sucrose is lower than that of acid catalysed hydrolysis.

B. During denaturation, secondary and tertiary structures of a protein are destroyed but primary structure remains intact.

C. Nucleotides are joined together by glycosidic linkage between C_1 and C_4 carbons of the pentose sugar

D. Quaternary structure of proteins represents overall folding of the polypeptide chain.

Choose the **correct** answer from the options given below:

(1) A, C and D Only

(2) A, B and D Only

(3) A and B Only

(4) B and C Only

62. The correct order of the rate of the reaction for the following reaction with respect to nucleophiles is:

$$CH_3Br + Nu^{\ominus} \longrightarrow CH_3Nu + Br^{\ominus}$$

(1) $PhO^- > {}^-OH > CH_3COO^- > ClO_4^-$

(2) $ClO_4^- > CH_3COO^- > {}^-OH > PhO^-$

(3) $CH_3COO^- > PhO^- > {}^-OH > ClO_4^-$

(4) ${}^-OH > PhO^- > CH_3COO^- > ClO_4^-$



63. Given below are two statements:

Statement I: Crystal Field Stabilization Energy (CFSE) of $[Cr(H_2O)_6]^{2+}$ is greater than that of $[Mn(H_2O)_6]^{2+}$.

Statement II: Potassium ferricyanide has a greater spin-only magnetic moment than sodium Ferrocyanide.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both **Statement I** and **Statement II** are true

(2) Both **Statement I** and **Statement II** are false

(3) **Statement I** is true but **Statement II** is false

(4) **Statement I** is false but **Statement II** is true

64. The correct increasing order of C – H(A), C – O(B), C = O(C) and C ≡ N(D) bonds in terms of covalent bond length is:

(1) A < B < C < D

(2) A < D < C < B

(3) D < C < B < A

(4) D < C < A < B

65. Given below are four compounds:

(a) n-propyl choride

(b) iso-propyl chloride

(c) sec-butyl chloride

(d) neo-pentyl chloride

Percentage of carbon in the one which exhibits optical isomerism is:

(1) 52 (2) 56

(3) 46 (4) 40

66. Given below are some of the statements about Mn and Mn_2O_7. Identify the correct statements

A. Mn forms the oxide Mn_2O_7 in which Mn is in its highest oxidation state.

B. Oxygen stabilizes the Mn in higher oxidation states by forming multiple bonds with Mn

C. Mn_2O_7 is an ionic oxide.

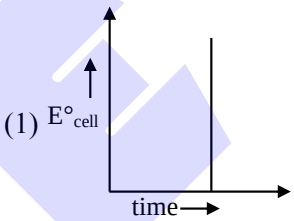
D. The structure of Mn_2O_7 consists of one bridged oxygen.

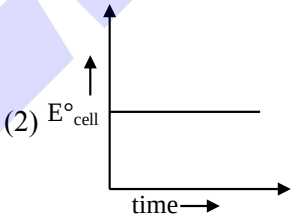
Choose the **correct** answer from the options given below:

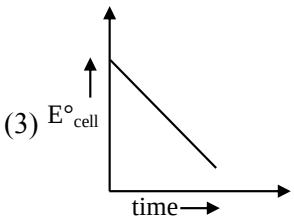
(1) A, B, C and D (2) A, B and D Only

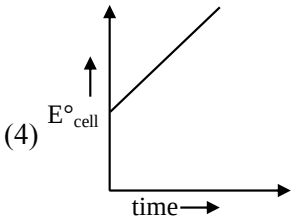
(3) A, C and D Only (4) A, B and C Only

67. For a closed circuit Daniell cell, which of the following plots is the accurate one at a given temperature ?

(1) 

(2) 

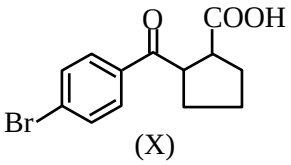
(3) 

(4) 



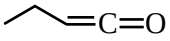
68. Given below are two statements :

Statement – I : Compound (X), shown below , dissolves in $NaHCO_3$ solution and has two chiral carbon atoms



(X)

Statement – II : Compound (Y), shown below, has two carbons with sp^3 hybridization, one carbon with sp^2 and one carbon with sp hybridization


(Y)

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement I** is true but **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

(4) Both **Statement I** and **Statement II** are false

69. Given below are two statements :

Statement I : The correct order in terms of bond dissociation enthalpy is $Cl_2 > Br_2 > F_2 > I_2$

Statement II : The correct trend in the covalent character of the metal halides is $[SnCl_4 > SnCl_2]$, $[PbCl_4 > PbCl_2]$ and $[UF_4 > UF_6]$

In the light of the above statements, choose the **correct** answer from the options given below :

(1) **Statement I** is true but **Statement II** is false

(2) Both **Statement I** and **Statement II** are true

(3) **Statement I** is false but **Statement II** is true

(4) Both **Statement I** and **Statement II** are false

70. On heating a mixture of common salt and $K_2Cr_2O_7$ in equal amount along with concentrated H_2SO_4 in a test tube, a gas is evolved. Formula of the gas evolved and oxidation state of the central metal atom in the gas respectively are :

(1) CrO_2Cl_2 and +5 (2) CrO_2Cl_2 and +6

(3) $Cr_2O_2Cl_2$ and +6 (4) $Cr_2O_2Cl_2$ and +3

SECTION-B

71. The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in 0.1 M H_2X solution is _____ $\times 10^{-15}$ M. (Nearest Integer)

72. The osmotic pressure of a living cell in 12 atm at 300 K. The strength of sodium chloride solution that is isotonic with the living cell at this temperature is _____ $g\ L^{-1}$. (Nearest integer)

Given : $R = 0.08\ L\ atm\ K^{-1}\ mol^{-1}$

Assume complete dissociation of NaCl

(Given : Molar mass of Na and Cl are 23 and 35.5 $g\ mol^{-1}$ respectively.)

73. A substance 'X' (1.5 g) dissolved in 150 g of a solvent 'Y' (molar mass = 300 $g\ mol^{-1}$) led to an elevation of the boiling point by 0.5 K. The relative lowering in the vapour pressure of the solvent 'Y' is _____ $\times 10^{-2}$. (Nearest integer)

[Given : K_b of the solvent = 5.0 $K\ kg\ mol^{-1}$]

Assume the solution to be dilute and no association or dissociation of X takes place in solution.



74. Identify the metal ions among Co^{2+}, Ni^{2+}, Fe^{2+}, V^{3+} and Ti^{2+} having a spin-only magnetic moment value more than 3.0 BM. The sum of unpaired electrons present in the high spin octahedral complexes formed by those metal ions is _____.

75. MX is a sparingly soluble salt that follows the given solubility equilibrium at 298 K

$MX(s) \rightleftharpoons M^+(aq) + X^-(aq)$; $K_{sp} = 10^{-10}$

If the standard reduction potential for

$M^+(aq) \xrightarrow{+e^-} M(s)$ is $\left(E^{\ominus}_{M^+/M}\right) = 0.79V$, then the value of the standard reduction potential for the metal/metal insoluble salt electrode $E^{\ominus}_{X^-/MX(s)/M}$ is _____ mV. (nearest integer)

[Given: $\frac{2.303RT}{F} = 0.059V$]



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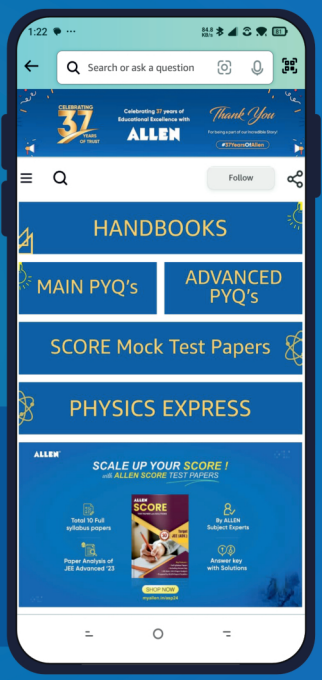
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